OPENAI SOLUTIONS ENGINEERING

From fragmented supply-chain signals to governed decisions

A proposed OpenAI platform approach for Zeiss Supply Chain Management and IT leadership

IN A NUTSHELL

TL;DR

OpenAI becomes the governed decision layer between ZEISS supply-chain data and operational action.

Executive chat experience

Enterprise integration and control layer

Problem

Context is split across SAP, logistics, suppliers and files.

Platform move

A central communication interface gives leaders **one decision surface**; the API integrates tools and controls.

Pilot

Start with one high-value workflow and validate faster triage, evidence quality and governed approvals.

Status quo and validation needs

Known operating context

- Advanced manufacturing, medical technology, quality and optics domains
- Decisions draw on SAP, supplier, logistics and document context
- Value depends on trustworthy answers, not another dashboard

Discovery hypotheses to validate

- Which disruptions consume the most time?
- Where do handoffs reduce accuracy or predictability?
- Which recommendations need evidence, review and traceability?

Providing a decision framework for existing data

SAP

- POs
- Material master
- MRP exceptions

Warehouse

- EWM stock
- Goods receipts
- Production buffers

Carriers

- DHL / FedEx
- ETA changes
- Customs events

Suppliers

- Commit dates
- Capacity
- Quality status

Files

- Scorecards
- Excel trackers
- Contracts

PAIN

- Manual effort to reconstruct one trusted view
- Answer accuracy varies by source snapshot
- Low precision for lead-time and supplier risk
- Reliability depends on scarce experts

COST

- Expedite spend and buffers absorb uncertainty
- Leadership time shifts into escalation meetings
- Late mitigation creates schedule churn
- Scattered evidence increases quality exposure

REFERENCE ARCHITECTURE

Solution overview

Supply Chain Hub

Conversation layer for asking, reviewing evidence and **aligning next steps**.

→

Intelligence layer

Applies GPT-5.6 model and reasoning controls to turn **questions into options**.

→

Operating controls

Server-side guardrails check prompt scope, enforce persona, workflow and selected-source access, filter financials and block unavailable actions.

→

ZEISS systems

SAP, logistics, supplier, warehouse and documents stay **sources of truth**.

Ground

Retrieve permitted, current supply-chain context.

Reason

Compare risk, options and trade-offs.

Act

Assign work or route accountable approval.

OpenAI Responses API

Streams source-backed answers with reasoning and model controls.

OpenAI Agents SDK

Runs persona-aware tools, assignments and reviewer handoffs.

Why OpenAI / ROI

Faster escalation cycles, less expedite exposure and a reusable business case.

EXAMPLE

Executive workflow

Trigger
Customer shipment at risk: supplier date, SAP availability and carrier ETA do not match

→

Question
Can we keep the promise date? What needs approval today?

→

Evidence
Orders, inventory, supplier commits, carrier ETA and expedite cost

→

Action
Recommended option, customer impact, owner and approval step

Before
A customer delivery is at risk, and the team spends the first 20 minutes **checking which source is actually current** instead of deciding what to do.

After
Leaders see the exposed orders, root cause, options, cost trade-off and approval owner **in one interface**.

Deployment constraints to validate

Data & privacy
Approved fields only; confirm GDPR, supplier confidentiality, retention and regional processing.

Identity & governance
SSO, server-side roles, masking and a named reviewer for every high-impact action.

Operations & adoption
Fail-closed fallback, process owner, training, feedback and security / works-council checkpoints.

Demo

Impact validation and success criteria

Validate a credible business case and earn a scale decision by testing real supply-chain scenarios with ZEISS business and IT owners.

1

Business value to validate

€180K–€310K

Illustrative annual gross value. The POC will confirm actual review volumes, expedite costs and disruption exposure against the ZEISS baseline.

€15K–€25K	€101K–€151K	€65K–€130K
Faster review work	**Fewer urgent expedites**	**Lower disruption exposure**
600–1,000 reviews × 20 minutes saved × €75/hour	12–18 avoided cases × €8,400	0.35–0.70 probability-weighted events × €185,000

Scale decision ZEISS confirms measured benefit and annual solution cost before rollout. Net ROI = (validated benefit − annualized solution cost) ÷ annualized solution cost.

2

How the POC earns a scale decision

1. **Agree realistic test scenarios.** Business and IT select routine cases, high-impact exceptions and permission boundaries.
2. **Compare each answer and proposed action.** Check the supporting evidence, serious-risk coverage, approved systems and required human approvals.
3. **Review failures together.** Correct the workflow and rerun the same scenarios before release.

Business value • Process owner

At least 25% faster review work and at least 80% user usefulness.

Decision quality • Supply-chain lead

At least 90% supported by approved sources and fewer than 5% of serious risks missed.

Technical reliability • IT / AI owner

At least 95% of runs use the correct approved systems and workflow.

Governance • Risk owner

Every sensitive or high-impact action receives accountable human review.

Time to value

WEEK 0

Align

Sponsor, workflow owner, success metrics, data boundaries.

WEEK 1

Connect

Representative sources, role scopes, approval policy.

WEEK 2

Build

App workflow, business tools, grounding and audit trail.

WEEK 3

Validate

User feedback, answer quality checks, security review.

WEEK 4

Decide

Impact readout, guardrail review, pilot recommendation.

After the POC

Weeks 5-8 · Pilot
Harden one workflow and onboard the first team.

Weeks 9-12 · Adopt
Embed ownership, feedback and support.

Next · Scale the blueprint
Reuse the pattern across Procurement, Quality, Manufacturing and Finance.

How OpenAI can support adoption

Solutions Engineering
Architecture, POC build and evals.

Enterprise support
Deployment readiness and escalation path.

Optional delivery services
Embedded help for production integration.

Annex

Behind the scenes

App experience

- Next.js App Router, React and strict TypeScript
- GPT-5.6 Sol, Terra and Luna with six reasoning levels
- Persona-aware tools, source settings, answer copy and feedback controls

Grounded chat and guardrails

- `/api/chat` runs prompt-scope checks before context, sources or tools
- Server context enforces role scope, selected sources and masked fields
- Responses streaming with deterministic workbook and demo fallbacks

Agentic actions and integrations

- `/api/actions` enforces persona, workflow and source eligibility
- Explicit assignee and reviewer metadata drives tasks and approvals
- Role-aware Microsoft 365 grouping, reviewer handoffs and trace flushing

Decision support and delivery

- Model-selected savings–relationship bubble or matrix; trusted records stay server-owned
- Server-derived decision colors align recommendations across both views
- Vitest, strict typecheck and OpenNext/Wrangler delivery to Cloudflare Workers